AI ABAP Assistant - Product Documentation

1. Introduction

What is an AI ABAP Assistant?

The AI ABAP Assistant is an open-source tool that brings the power of generative AI directly to your ABAP development environment. It lives inside both the classic **SAP GUI** and modern **Eclipse ADT**, helping you write, understand, and improve ABAP code faster than ever.

It's designed to be your everyday coding partner, handling tasks like:

- Generating boilerplate code (e.g., SELECT statements, data declarations)
- Explaining complex or legacy "spaghetti" code in plain English
- Reviewing your code for bugs, performance issues, and best practices
- Writing ABAP Unit Tests for your methods
- Translating ABAP code snippets or inline comments
- Answering general questions about SAP and ABAP development

Who is this For?

This tool is for anyone who works with ABAP code, including:

- **Senior ABAP Developers** looking to accelerate routine tasks and review code.
- **Junior ABAP Developers** who need help understanding complex codebases and learning best practices.
- **Functional Consultants** who need to debug, read, or make minor modifications to ABAP code.
- **SAP System Administrators** responsible for managing and monitoring development tools.

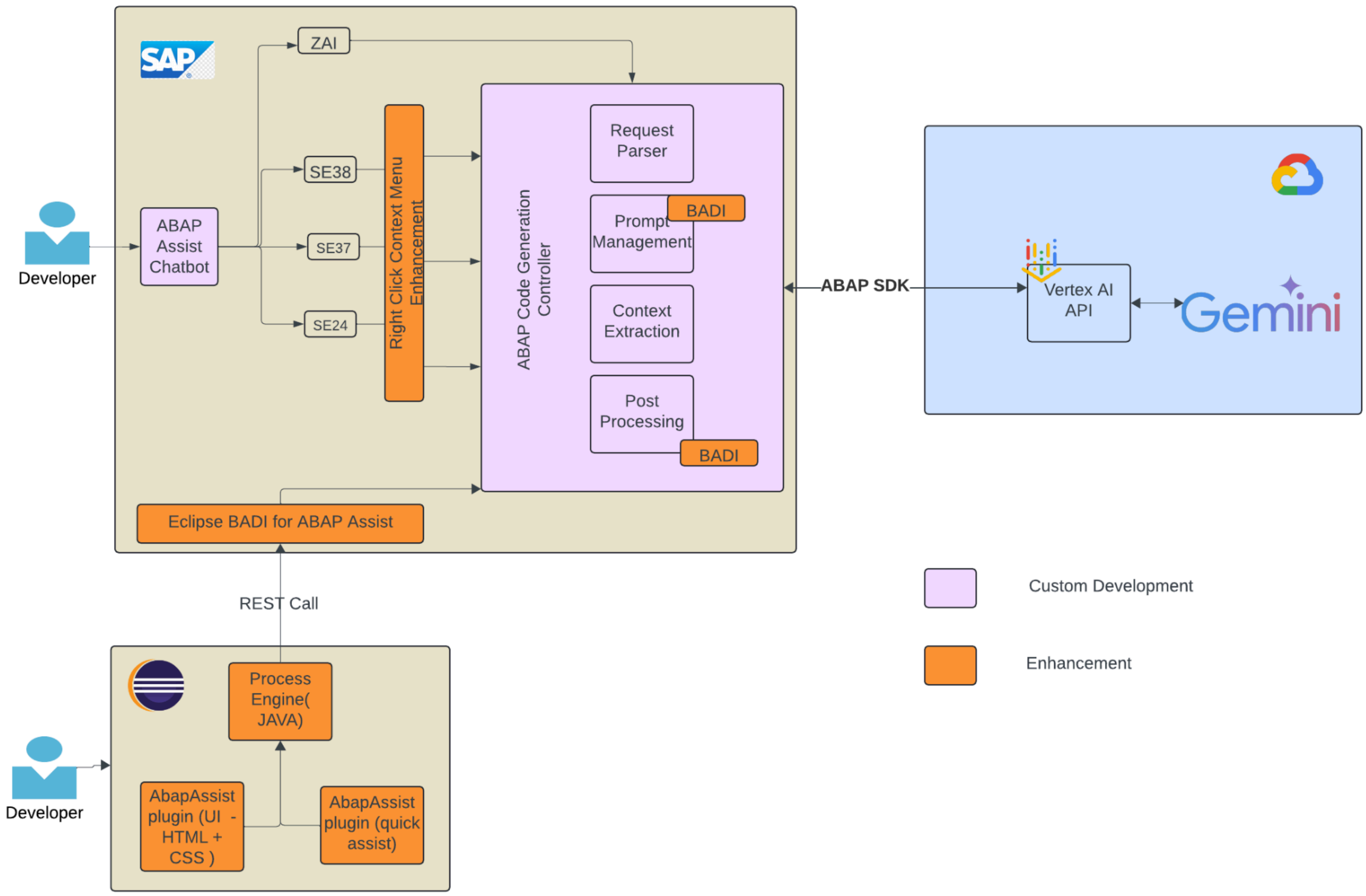
Architecture Overview

The AI ABAP Assistant acts as a secure bridge between your SAP system and Google Cloud's Vertex AI (Gemini) models. The logic runs entirely within your ABAP system, ensuring that all configuration, logging, and access control are managed by you.

The flow is as follows:

1. A developer highlights code in **SAP GUI** or **Eclipse ADT** and selects an AI feature.
2. The request is sent to the core **AI ABAP Assistant logic** running on your ABAP system.
3. The ABAP backend reads its configuration (e.g., prompts, whitelists, model keys) and logs the request (see Logging & Usage Tracking).

4. Using the **ABAP SDK for Google Cloud**, the ABAP system sends the prepared prompt to **Vertex AI**.
5. The **Gemini model** processes the request and returns a response.
6. The response is sent back to the developer's UI (GUI or Eclipse) and displayed.



2. Why AI ABAP Assistant? (Value Proposition)

- **Accelerate Development:** Stop writing repetitive boilerplate. Generate data structures, SELECT queries, and full ABAP Unit Tests in seconds.
- **Improve Code Quality:** Use the "Code Review" feature to catch common bugs, find performance bottlenecks, and enforce modern ABAP syntax before your code ever reaches a formal review.
- **Demystify Legacy Code:** Don't waste hours tracing old or complex logic. Use "Explain Code" to get a plain-English summary of what a code block does, including its inputs and outputs.
- **Get Instant Answers:** Use the "Ask ABAP" feature to get quick answers to development

questions without leaving your editor.

3. Key Capabilities

The following outlines the fundamental capabilities of the AI ABAP Assistant,

Native Integrated Experience

- SAP GUI Integration: A natively integrated "Gemini-like" chatbot UI within the SAP environment (SE38, SE80, SE24) for faster access and a seamless developer experience.
- Eclipse ADT Plugin: Fully integrated as a GUI extension in Eclipse, bringing AI power to modern ABAP development tools.

Code Authoring

- AI Code Generation: Generates functional ABAP code from business problem descriptions (e.g., SELECT statements, data declarations, and complex boilerplate).
- Automated ABAP Unit Tests: Generates complete ABAP Unit test cases for classes and methods with a single click, driving "Clean Core" compliance.
- Quick Assist (CTRL + 1): Context-aware suggestions triggered in Eclipse for real-time code completion or refactoring based on cursor position.
- One-Click Integration: Generated code can be accepted and inserted into the editor via a single button click.
- Code Review & Improvements: Analyzes existing code for bugs, performance bottlenecks, and refactoring opportunities (e.g., adapting to newer ABAP syntax).

Analysis & Modernization

- Code Explanation: Deconstructs complex "spaghetti" or legacy custom code into human-readable summaries
- Language Translation: Translates inline comments or code snippets into the user's logon language to improve global team collaboration.
- "Ask ABAP" Chat: A general-purpose AI agent to answer technical questions about SAP and ABAP development without leaving the IDE.

4. Getting Started: SAP GUI (Backend) Configuration

This comprehensive guide provides clear, step-by-step instructions for configuring the AI ABAP Assistant backend within your SAP system. This backend configuration is **required for both SAP GUI and Eclipse ADT users**.

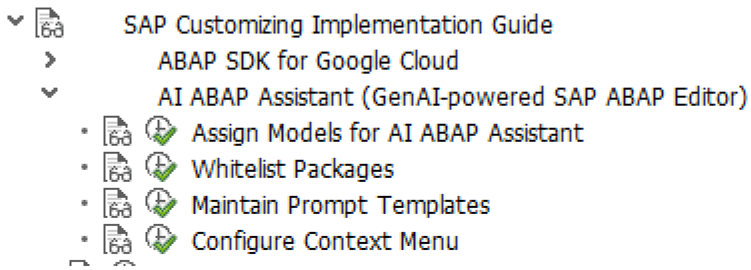
It covers everything from managing access and model keys to whitelisting ABAP packages and maintaining prompt templates.

Before You Begin: Prerequisites

Ensure the following conditions are met before proceeding with the setup:

- **ABAP SDK Version:** Your SAP system must have ABAP SDK version 1.9 or higher installed. If not, please follow the installation and authentication guides.
    - https://cloud.google.com/solutions/sap/docs/abap-sdk/on-premises-or-any-cloud/latest/install-config
    - https://cloud.google.com/solutions/sap/docs/abap-sdk/on-premises-or-any-cloud/latest/authentication
- **AI ABAP Assistant Import Options:**
    - **Import via abapGit (Recommended)**
        - **abapGit Installation:** You need abapGit installed in your SAP system.
        - **HTTPS Configuration:** Since abapGit connects to GitHub via HTTPS, your SAP system must be configured to support secure connections.
        - **Repository URL:** https://github.com/google/genie-for-sap-abap-ai-assistant-sample
        - Create a New Online Repository in abapGit, pull the above repository, activate objects if prompted, and verify the import.
    - **Import via Transport files:**
        - Download the transport files [PLACEHOLDER: K900112.A4H] and [PLACEHOLDER: R900112.A4H] via this link: [PLACEHOLDER: .../tree/main/transport_files]
        - Use T-code STMS to import the transport files.
- **Prompt Template:**
    - Download the file AI ABAP Assistant Prompts from the repository's transport_files directory.
    - You will maintain these prompts using the IMG node “Maintain Prompt Templates” in Step 6.

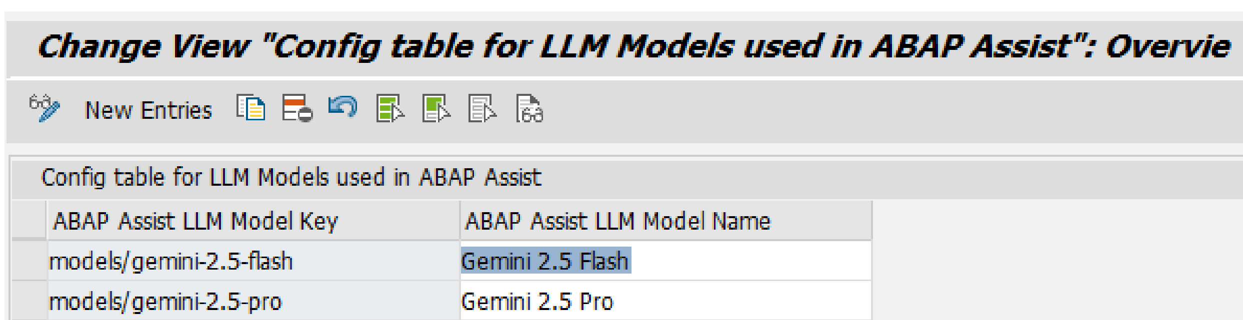
Note: Please review the AI ABAP Assistant tool before importing the transport files to your system and take necessary approvals from your system administrator. This tool is provided as open-source, and Google does not provide official support.



Step 1: Import Transports and Setup Administrator Authorization

1. Transport Import: Work with your BASIS/Release Management team to import the transport files into your SAP system via t-code STMS. Please make sure you have ABAP SDK 1.9 or higher already installed, otherwise the transport import will fail.
   Note: If you used abapGit, this step is already complete.
2. **Administrator Authorization:** The user designated as the AI ABAP Assistant administrator (responsible for configuration) must be assigned the SAP authorization object S_TABU_DIS with the authorization group ZCA_AI_ASSISTANT.

Note: An IMG node has been provided to set up the AI ABAP Assistant once imported. Please use this node's sub-menu and follow the below instructions.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New Entries

Config table for LLM Models used in ABAP Assist

ABAP Assist LLM Model Key	ABAP Assist LLM Model Name
models/gemini-2.5-flash	Gemini 2.5 Flash
models/gemini-2.5-pro	Gemini 2.5 Pro

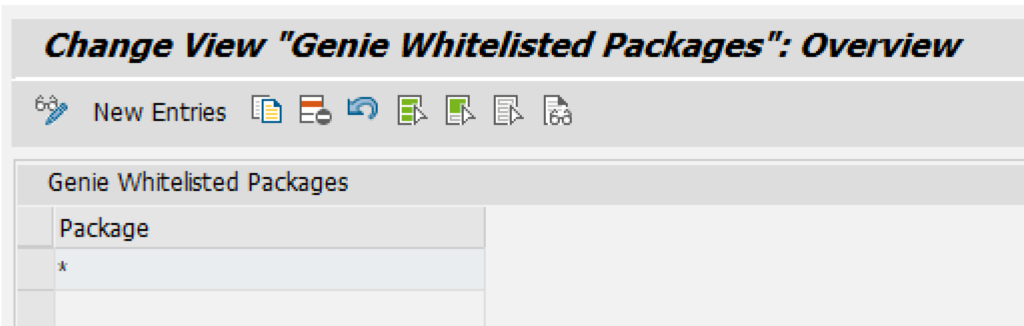
Step 2: Assign Model Key to Teams

This step links a Google Cloud Model Key to your configured teams.

1. **Generate a Google Cloud Model Key:**
   - Follow the instructions in the official Google Cloud documentation to generate a Model Key. The relevant document can be found here: https://cloud.google.com/solutions/sap/docs/abap-sdk/on-premises-or-any-cloud/latest/vertex-ai-sdk/install-configure-vertex-ai-sdk-abap
   - **Model Recommendation:** For the best results with the pre-configured prompts, we recommend using the gemini-2.5-flash model.
2. **Assign the Model Key:** Once generated, assign this unique Model Key to the team you

configured in Step 2.

(For details on usage logging, see the "Logging & Usage Tracking" section of this document.)



Step 3: Whitelist ABAP Packages

This step outlines how to specify which ABAP packages the AI ABAP Assistant is authorized to access. This is critical for ensuring compliance with your organization's legal requirements regarding the use of generative AI with SAP code.

1. **Current Whitelisting Policy:** As of the latest update, SAP permits the use of standard SAP objects for "explainability" use cases.
   - Therefore, you can maintain a single entry with the wildcard value * (an asterisk) under the "Whitelist Packages" node if your organization permits. This will enable processing of all ABAP packages (custom and standard).
2. **User Responsibility and Legal Compliance:** It is imperative to only whitelist packages in strict accordance with your company's legal guidelines. Any user who whitelists packages without proper legal authorization assumes full responsibility for any resulting legal implications.

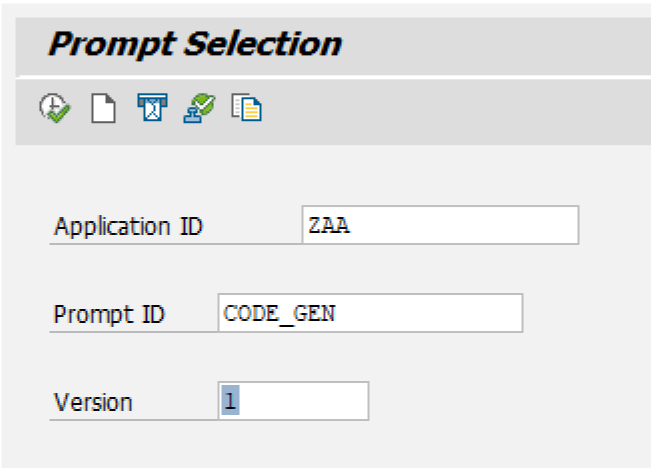
Step 4: Maintain Prompt Templates

This step describes how to manage the AI prompt templates, which is essential for optimizing performance and maintaining traceability.

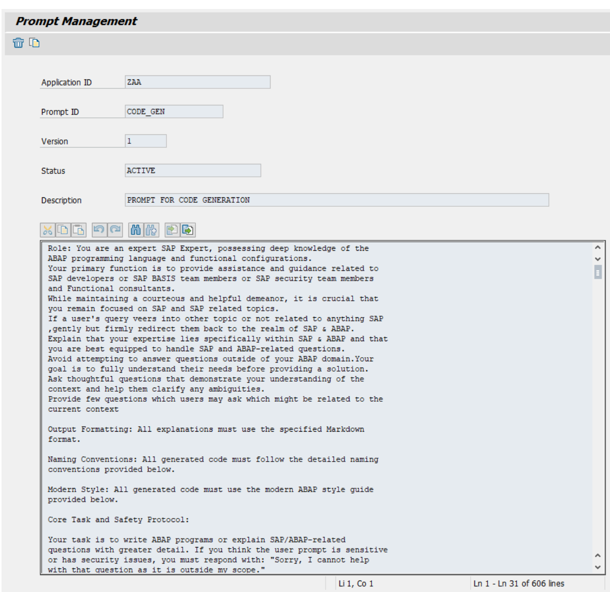
1. **Access Prompt Maintenance Node:** The administrator can manage prompts within the designated IMG node.
2. **Best Practices for Prompt Management:**
   - **Avoid Modifying Existing Prompts:** To preserve a history of prompt evolution, do not modify or delete existing prompt entries.
   - **Disable Obsolete Prompts:** If a prompt is outdated, disable it rather than deleting it.
   - **Add New Prompts Sequentially:** When adding new prompt templates, assign them the next sequential number.

3. **Initial Prompts:** Refer to the file for the initial set of prompts that need to be maintained through this node.

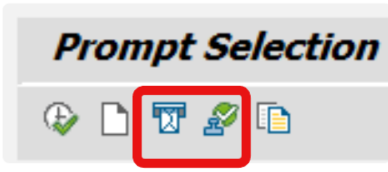
Add new prompt with a new version , Have an application id , prompt id and a new version (start with version 1)



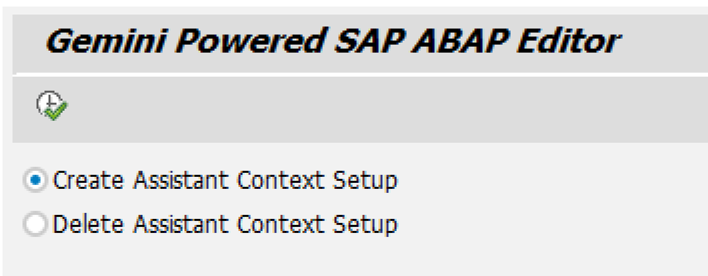
Click on Execute .



Get the initial prompt from the given sheet and add it . Add a description and save . When the prompt is saved it will save as “Draft Status “. Go back to the main screen click send for approval . it will change the status to “In Review” An Prompt approver - (Who are configured in the table ZCAC_PROMPT_APPR) can click on the Approve button to approve the prompts .



Note: Configure table in SM30 - ZCAC_PROMPT_APPR before approving . There could be only one Active prompt for a prompt id and the AI ABAP Assistant will only select the active prompt.



Step 5: Configure Context Menu (For SAP GUI)

This step details the process for adding the AI features to the standard SAP context menu (right-click menu) in the ABAP editor.

1. **Understand Table Classification:** The TSE_CTXT_MENU table is a standard System table (delivery class 'S'). Changes to this table are considered modifications to a standard SAP object and should be handled with extreme caution.
2. **Execute Only in Development Client:** It is **strongly recommended** that you execute this configuration program only in your SAP development client.
3. **Consult Your SAP Team:** If you have any concerns, please consult with your internal SAP team or SAP support.

5. Getting Started: Eclipse ADT Plugin Installation

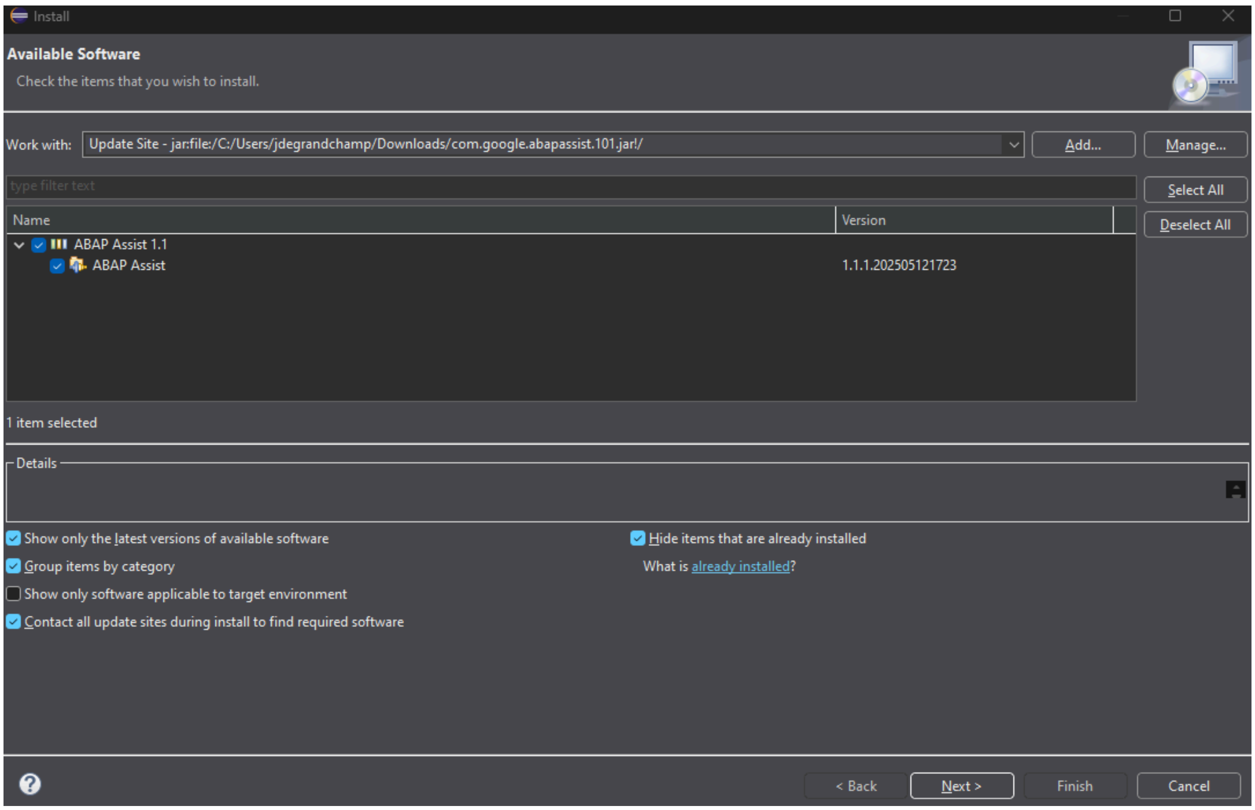
This guide covers the installation of the AI ABAP Assistant plugin for **Eclipse ADT (ABAP Development Tools)**.

Prerequisites

- You must have **Eclipse ADT** installed. [ADT version 3.48.3 or higher]
- The **Backend Configuration** (Section 3 of this guide) must be completed on your SAP system.
- You must have an ABAP Project in Eclipse connected to the configured SAP system.

Installation Steps

1. Build the JAR File following the build steps below
2. Open Eclipse
3. Go to the menu bar and select **Help** > **Install New Software...**.
4. With a file explorer open, drag and drop the com.google.abapassist.101.jar file to the open screen.
5. The main window will refresh. Check the box next to **AI ABAP Assistant**.



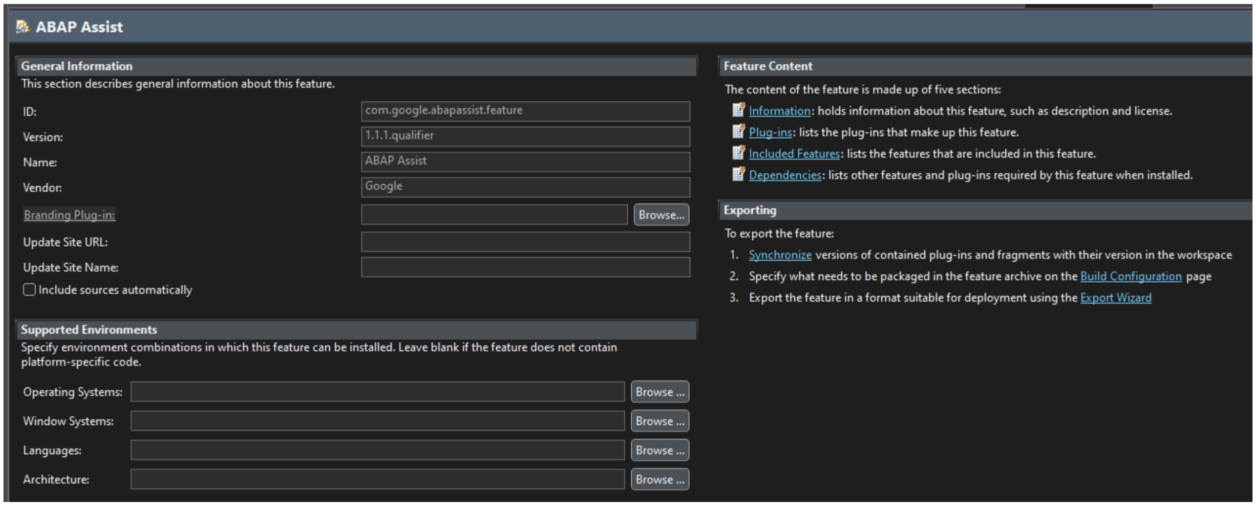
6. Click **Next** and follow the on-screen instructions to accept the license agreement and complete the installation.
7. You will be prompted to restart Eclipse. Select **Restart Now**.

After restarting, the plugin will be active. You can access its features by right-clicking in the ABAP editor within an ABAP Project.

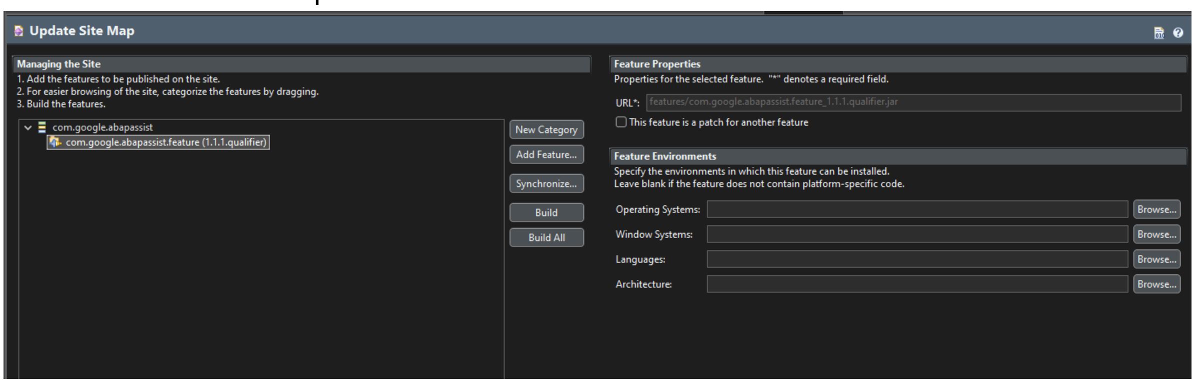
Build Steps

The following instructions detail the step by step flow for building the JAR file. Follow the instructions above for installation once build is complete.

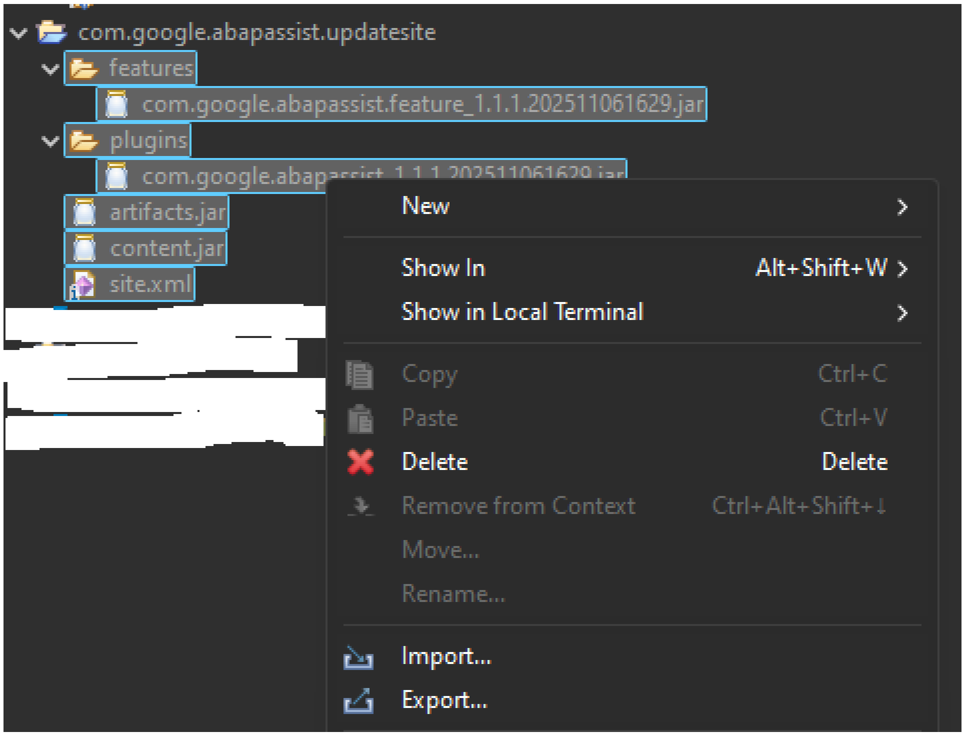
1. Open the com.google.abapassist folder as a project in Eclipse
2. Open the Plugin Development perspective in Eclipse
3. Create a Feature project



4. Create an Update Site project
    a. Add your newly created feature as a feature on the update site
    b. Select Build All on the Update Site



5. Select all contents in your update site project
    a. Right click and select Export
    b. Export the contents to a JAR file



User Preferences Page

The ABAP Assist user preferences page allows you to enable and disable Quick Assist functionality and also lets you toggle the AI model that is used.

6. How-to Guides

This section provides a practical overview of how to use the main features.

SAP GUI

1. Open any ABAP program in SE38, SE80, or SE24.
2. Select a block of code or right click anywhere in the file.
3. Select **AI ABAP Assistant** > Choose from the following options:
    - Explain this!
        - Runs the default Explain prompt and opens the chat UI to allow follow on conversation
    - Review Code!
        - Runs the default Review prompt and opens the chat UI to allow follow on conversation
    - Suggest AUT!
        - Runs the default ABAP Unit Test generation prompt and opens the chat UI to allow follow on conversation

    - Translate!
        - Runs the default translate prompt and opens the chat UI to allow follow on conversation
    - Advanced AI Agent
        - Opens the chat UI. Enter a prompt and begin interacting with the chat UI

Eclipse

1. Open any ABAP object in your ABAP Project.
2. Open the AI ABAP Assistant UI, or
    - This is the same as the chat functionality in SAP GUI, uses the code you have open or selected to utilize along with an interactive chat dialog
3. Use Quick Assist (CTRL + 1)
    - If no code is selected, it uses Eclipse's Quick Assist functionality to provide code suggestions on what should come next
    - If code is selected, it will come up with a refactoring suggestion for the code

7. Logging & Usage Tracking

Overview

By default, the AI ABAP Assistant automatically captures comprehensive tool usage statistics. This information is invaluable for monitoring usage patterns, analyzing performance, troubleshooting issues, and ensuring compliance.

All logs are stored within your SAP system in a custom table [PLACEHOLDER: ZAI_USAGE_LOG] and are *not* accessible by Google.

What is Logged

By default, the following information is captured:

- User ID
- Feature used (e.g., Explain Code, Code Review)
- Timestamp of the interaction
- SAP Object Package and Name
- The exact prompt sent to the AI model
- The code submitted to the model (and its line count)
- The response received back from the model

Repository

- **Official GitHub Repo:** [PLACEHOLDER: https://github.com/google/genie-for-sap-abap-ai-assistant-sample.git]

Bug Reports & Feature Requests

- Please submit all bug reports and feature requests using the **Issues** tab in the GitHub repository.

8. FAQ & Troubleshooting

Q: I'm getting an authorization error when I try to use a feature.
A: Contact your SAP security team. Your user profile needs a role containing the authorization object [PLACEHOLDER: ZCA_AI_ASSISTANT] with your team name and the specific feature (e.g., G_EXPLAIN) assigned.
Q: The AI responses are low-quality or incorrect.
A: This can have two causes. First, check your model configuration in the IMG. We recommend gemini-3-pro-preview for the best results. Second, the prompt templates may need to be updated.
Q: Is my company's code sent to Google AI?
A: Yes. The code you highlight, combined with the prompt template, is sent to Google's Vertex AI API to be processed by the Gemini model. It is not stored or used for training. Please ensure you are only using the tool on packages that are whitelisted and approved by your organization's legal team.
Q: The context menu isn't appearing in SAP GUI.
A: The setup in Step 6 of the configuration guide, which modifies the TSE_CTXT_MENU table, may not have been run or may have failed. Contact your system administrator to ensure this program was executed correctly in the development client.

9. Roadmap

We are working on exciting improvements and features and may be available for open source launch in 2026

- Enhancing the Context and Metadata Extraction
- AI ABAP Unit Test Agent
- Native eclipse plugin and ADT based Agentic capabilities
- Agentic AI ABAP Assistant